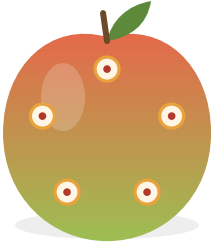


AN ERROR BUDGET FOR NIR FRUIT CALIBRATION

reference-replicate design rule · falsifiable attenuation diagnostic · leakage-audited validation

apples 633 × 5 faces · kiwifruit 4,318 × 2–5 devices

1 REFERENCE SIDE



- apples × flesh faces
633 × 5
- median 5-face range
2.80 °Brix
- within-fruit SD vs meter
5.6 × spec (±0.2)

one fruit, five answers

2 ERROR BUDGET

$$Y_{ij} = \mu + a_i + e_{ij}$$

$$\sigma a^2 = 2.327 \quad \sigma f^2 = 2.551$$

reference variance

ICC 0.477 [0.436, 0.515]

52.3% within

47.7% between

$$RMSEP^2 = model + \sigma f^2/m + instrument$$

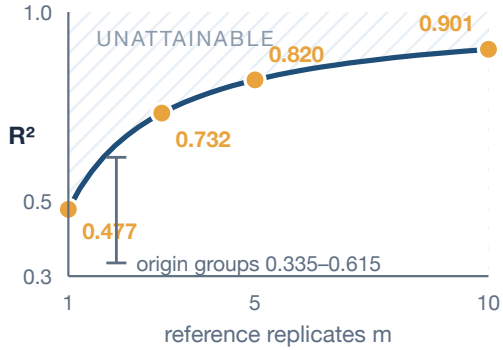
model

reference

instrument

two of the three parts are not the model's —
decompose before tuning models

3 ATTAINABLE CEILING



4 DESIGN RULE

$$m \geq (\sigma f^2 / \sigma a^2) \cdot R^2 / (1 - R^2)$$

target R^2

faces m ≥

0.70

3

0.80

5

0.90

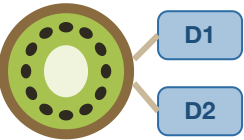
10

0.95

21

$\sigma f^2 / \sigma a^2 = 1.0965$ from a pilot batch · add one extra face as margin

5 INSTRUMENT SIDE (KIWIFRUIT)



SNV-corrected spectral variance

52.6%

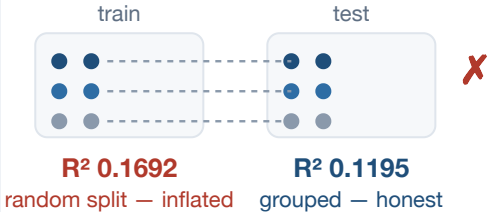
17.1%

30.3%

between fruit · device · fruit×device

device-linked error ≈ 0.505 %DM — spans just 20.6% of its mean across ten points
yet its share of MSE rises 4.3% → 24.7% (×5.7) as the model improves

6 VALIDATION AUDIT — SPLIT LEAKAGE



random splitting inflates R^2 by 1.41–1.46×

7 FALSIFIABLE ATTENUATION DIAGNOSTIC

six parameter-free predictions — all confirmed on held-out data

label-only baseline R^2



pred 0.3462
meas 0.3456



attenuation ratio



pred 0.5816
meas 0.5636



REPORTED R^2 MUST BE READ AGAINST A REFERENCE-LIMITED CEILING, NOT AGAINST 1 — THE CEILING IS COMPUTABLE BEFORE THE EXPERIMENT